

Shadow-Loom: An Experimental Framework for Combining Graphical Causal Reasoning with a Game-Engine-Style Graphical World Model of Narrative

David Wilmot

david.wilmot@gmail.com

github.com/davidrwilmot/shadow-loom

Released under AGPL-3.0-or-later + Non-Commercial Clause

Abstract

Stories hold a reader’s attention not, on the whole, because they are sequences of likely next words, but because they have causes, secrets, and consequences. **Shadow-Loom** is an experimental open-source framework that takes a narrative and turns it into a typed, versioned *graphical world model*. Two engines act on that model — both, on this account, *graphical world models* in their own right, since each defines a typed dynamics over the shared graph — in something of the spirit of a game engine that pairs a physics simulator with a higher-level director. The first engine is a *causal physics* (a Pearl-style graphical world model of cause and effect) grounded in Pearl’s ladder of causation (Pearl, 2009; Pearl and Mackenzie, 2018; Bareinboim et al., 2022) and the recently proposed counterfactual calculus over Ancestral Multi-World Networks (Correa and Bareinboim, 2025). The second is a *narrative physics* (a graphical world model of reader affect) that scores the same graph against four structural reader-states — mystery, dramatic irony, suspense (Wilmot and Keller, 2020), and surprise (Itti and Baldi, 2009) — in the tradition of Sternberg’s curiosity / suspense / surprise triad (Sternberg, 1992), and plays something like the role of the drama manager in classical interactive-narrative architectures (Mateas and Stern, 2003; Riedl and Bulitko, 2013; Ware and Young, 2014). Large language models are used only at the boundary: extraction, rendering, and audit. Identification, intervention, and counterfactual reasoning are carried out in typed code over the graph, in much the way a game engine restricts a stochastic agent to physically admissible actions (Ha and Schmidhuber, 2018; Park et al., 2023). The same loop — ingest, simulate, generate, audit, re-ingest — runs over user prose and produces a versioned world model with explicit, inspectable mathematical structure.

Two motivations sit behind the design. First, counterfactual reasoning sits on the rung of

Pearl’s hierarchy where current LLMs are weakest. They are reasonably competent at observational and even interventional causal questions in text (Kıcıman et al., 2024), but they tend to fail to act on the implicit beliefs and counterfactual states they themselves infer (Gu et al., 2026; Kim et al., 2023; Xu et al., 2025), and the foundations of rung-3 reasoning appear to need dedicated graphical machinery (Correa and Bareinboim, 2025; Bareinboim et al., 2022; Halpern, 2016). Second, coding agents tend to outperform open-domain agents at fixed model capacity, presumably because their outputs can be compiled, executed, and tested against a typed substrate. Shadow-Loom attempts to give narrative reasoning the same affordance: candidate continuations are propagated through the graph, checked against counterfactual-calculus-style independence and exclusion rules (Correa and Bareinboim, 2025), and re-ingested for round-trip verification (Madaan et al., 2023; Bai et al., 2022; Gu et al., 2024).

The system is offered as a research artefact rather than as a benchmarked NLP model. Formal evaluation is left to future work. The modest claim made here is that wiring a causal-graphical substrate underneath a constrained-LLM renderer is a tractable design point, with some apparent relevance to controllable generation, narrative theory of mind, computational social science, and the digital humanities. Code, fixtures, and pipeline are released open source in the hope that other researchers — in NLP, in computational social science, and in the digital humanities — may find the substrate useful and adapt it to their own questions. Definitions and equations for each pipeline stage are given in the appendix.

1 Introduction

Most current LLM-driven story systems treat a narrative as a sequence-prediction problem and absorb the causal, temporal, and epistemic structure of the story into the weights of a single model. Recent work shows that such models are competent at *read-*

ing causal relations from text (Kıcıman et al., 2024) but markedly weaker at *acting* on the implicit beliefs of characters they have just inferred (Gu et al., 2026; Kim et al., 2023; Cross et al., 2024); in unconditioned generation they tend to collapse onto a small set of shared plot tropes (Xu et al., 2025; Tian et al., 2024). The complementary direction explored here keeps an explicit, typed graph of the storyworld (Schank and Abelson, 1977; Mani, 2012; Elson, 2012) and performs identification, intervention, and counterfactual reasoning over it symbolically (Pearl, 2009; Halpern, 2016; Correa and Bareinboim, 2025), using the LLM only as a constrained renderer (Riedl and Young, 2010; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Yang et al., 2023). Concretely, a query like “what if Macbeth refuses to murder Duncan?” becomes a rung-2 do-operation on a typed event graph extracted from the play (App. B.2) rather than a free-form prompt to a language model; “does Poirot know the conspirators are still lovers?” becomes a finite belief-set membership test (App. B.4) rather than a single forward pass.

Motivation 1: counterfactuals are where LLMs tend to struggle. Pearl’s hierarchy is provably non-collapsible. Rung-3 quantities are not, in general, identifiable from rung-2 data, let alone from rung-1 observation (Bareinboim et al., 2022). Empirically, LLMs reproduce surface causal patterns from their training corpus (Kıcıman et al., 2024) but tend to degrade sharply on multi-hop belief-conditioned action prediction (Kim et al., 2023; Gu et al., 2026) and on story-internal counterfactual consistency (Xu et al., 2025). The questions that ordinarily matter to a reader of a story — “what if Macbeth had refused?”, “what if Roy had told Mary the truth?” — live on the rung where the modern foundations (AMWN + ctf-calculus, Correa and Bareinboim, 2025) are designed to operate, and where end-to-end neural generators are not. A typed graph plus an explicit do-operator and abduction step turns each such question into a finite sequence of well-defined operations (Pearl, 2009; Halpern and Pearl, 2005; Halpern, 2016) rather than a free-form imagination task.

Motivation 2: coding agents do better, arguably, because they compile and test. A familiar pattern in agent design is that coding agents tend to outperform open-domain agents at fixed model capacity. Their outputs can be compiled, executed against a test suite, and rejected when they fail.

The lever, on this reading, is the *typed substrate underneath the natural-language reasoning*, not the model itself. Shadow-Loom borrows that lever for narrative. A continuation is rejected if (a) its causal propagation (Eq. 1) violates inertia or affordance gating; (b) its counterfactual branch fails an AMWN-style consistency, independence, or exclusion check (Correa and Bareinboim, 2025); or (c) its re-extraction does not match the brief that licensed it (the audit pass, in the LLM-as-judge tradition of Bai et al., 2022; Madaan et al., 2023; Gu et al., 2024). The graph plays something like the role of the compiler; the auditor plays something like the role of the test suite.

The framework combines four ideas which, so far as I can tell, have not previously been wired together in a single open-source pipeline:

1. A **Pydantic-typed world graph** (WorldStateV1) with entities, events, beliefs, locations, channels, and three kinds of edge (causal, social, spatial), every node and edge tagged with both a chronological *fabula* time and a narrative *syuzhet* index (Shklovsky, 1965; Genette, 1980).
2. A **causal physics engine** — a graphical world model of cause and effect — that implements Pearl’s three rungs — observation, do-intervention, and abductive counterfactual — with a propagation rule gated by an explicit “Impact > Inertia” threshold and spatial affordance checks.
3. A **narrative physics layer** — a graphical world model of reader affect over the same graph — that scores it against closed-form information-state functionals for mystery, dramatic irony, suspense, and surprise. Suspense extends the hope/threat formulation of Wilmot and Keller (2020) as a *balance* $\times$ *stakes* product over the unrevealed ledger; surprise is binary KL divergence between a leave-one-out corpus-marginal prior, evidence-updated by revealed causal edges, and the entity’s reconstructed final state.
4. A **constrained-LLM renderer + recursive auditor** loop (Gu et al., 2024) that converts the simulation result into a typed creative brief, generates prose under that brief, and then re-extracts the prose to verify no “miracle steps” were introduced.

The system is named for the dual-graph metaphor: a *loom* of factual edges runs in parallel with one or more *shadow* counterfactual branches, in the sense of the AMWN construction of [Correa and Bareinboim \(2025\)](#). Section 2 sketches the pipeline; Section 4 states what is novel and what is borrowed; Section 5 gives the design rationale; Section 6 discusses relevance to NLP, reasoning, and social science; Section 7 states the limitations honestly. Equations and per-stage definitions are in App. A; an extended narrative walkthrough of the whole loop on a single fixture is in App. C; worked examples drawn from the sixteen bundled plot fixtures follow in App. B; and a separate appendix on the authoring user interface, with screenshots and step-by-step example sessions, is in App. D.

How to read this paper. The main text is intentionally short. It states the design, situates it in the literature, and records what is novel as a combination. Readers who want to understand the symbolic machinery in detail — the schema, the ego-graph extraction, the do-operator, the abductive counterfactual, the four affective scorers, the brief-and-audit loop — should treat App. A as the authoritative specification. App. C is a long-form description of the same pipeline expressed as a story: *Macbeth* is fed in as raw prose and followed step-by-step through ingestion, intervention, propagation, scoring, rendering, and audit, with every intermediate object shown. The eight example sub-appendices in App. B then show how each individual stage manifests on a different bundled fixture. App. D is for readers who want to understand or use the authoring workspace; it cross-links every UI surface to the pipeline stage it exposes and includes example sessions on real fixtures.

2 Pipeline overview

The pipeline is a five-phase loop over a versioned world model. Every version is ancestor-linked, so a counterfactual fork is literally a sibling row in a directed acyclic version tree, not a mutation of canon. Figure 1 sketches the loop.

The five phases correspond to the five appendix subsections (A.2–A.7): (1) **Ingestion** runs five LLM agents to extract a typed `GlobalRegister`, then three concurrent specialist agents (physics, social, consequences) per chunk to extract events, relationships, channels, and entity-state deltas, with a programmatic validation and auto-repair loop. (2) **Causal physics**

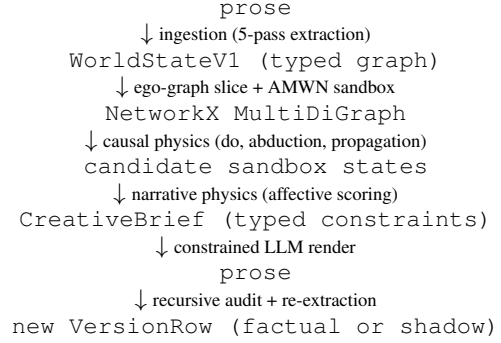


Figure 1: The Shadow-Loom loop. The LLM is invoked only at the top (ingestion), the bottom (rendering), and the audit; everything in between is symbolic computation over the graph. See the appendix for definitions of each stage.

forks an Ancestral Multi-World Network sandbox, applies a do-operator for interventions, runs abduction for counterfactuals, and propagates effects through a topological sort of the causal sub-graph subject to inertia and affordance gating. (3) **Narrative physics / affective calculus** scores each candidate sandbox against four structural information-state functionals and six trait-trajectory emotional functionals. (4) **Generation** packages the highest-scoring candidate as a `CreativeBrief` of typed `ConstraintBlocks` and hands *only* the brief to a creative LLM. (5) **Audit and feedback** runs an LLM-as-judge over the prose, detects “miracle steps” (state changes with no licensing edge in the brief), and either accepts the version or re-runs generation with the auditor’s feedback appended, up to a configurable retry budget.

The entire loop is exposed both through a NiceGUI authoring workspace (eight cross-linked tabs over a versioned project DAG; App. D) and through a Model-Context-Protocol server with roughly forty tools, so a downstream agent can drive shadow-branch creation, scoring, and promotion programmatically.

3 Theoretical lineage and computational counterparts

The system has been built theory-first: every named field in `WorldStateV1` corresponds to a specific narratological, cognitive, or causal-inference construct, and every field has a prior computational instantiation that the design either generalises or borrows from. The pairings the design depends on are summarised below; an extended mapping lives in the project’s academic-foundations document.

Fabula vs syuzhet (Shklovsky, 1965; Tomashevsky, 1965; Genette, 1980; Bordwell, 1985; Bal, 2009). Two integer indices on every `EventNode`: chronological $f(e)$ and presentation $s(e)$. The four affective scorers reduce to set operations between the two projections (App. A.5).

Possible-worlds narratology (Ryan, 1991, 2001; Doležel, 1998; Pavel, 1986), modal-logical roots (Lewis, 1973), and counterfactual graphical models (Balke and Pearl, 1994; Shpitser and Pearl, 2008; Correa et al., 2021; Correa and Bareinboim, 2025). Every node and edge carries `world_id` $\in \{\text{factual, shadow}\}$; counterfactual queries fork a sibling `VersionRow`. The AMWN tag and three-rule mental model (consistency, independence, exclusion) come from Correa and Bareinboim (2025); I adopt the naming and architecture even though the typed narrative graph used here is not a rigorous SCM. AMWN supersedes the Twin-Network construction (Balke and Pearl, 1994) and the multi-network machinery of Shpitser and Pearl (2008), and is the counterfactual-completeness counterpart to do-calculus (Pearl, 1995, 2009; Bareinboim et al., 2022).

Greimas (Greimas, 1983), Propp (Propp, 1968), Bremond (Bremond, 1973), and the symbolic story-grammar tradition (Rumelhart, 1975; Mandler and Johnson, 1977; Kintsch and van Dijk, 1978; Schank and Abelson, 1977; Lehnert, 1981; Meehan, 1977; y Pérez and Sharples, 2001; Bringsjord and Ferrucci, 2000; Chambers and Jurafsky, 2008). `GlobalTrait` (`WORLD_`) realises Greimas’s “Power” actant; `RelationshipEdge.power_dynamic` mirrors the sender / receiver / helper / opponent topology; `EventNode.event_type` $\in \{\text{choice, outcome, revelation, utterance}\}$ is a coarse generalisation of Propp’s functions, closer in spirit to Bremond’s triad. The plot-unit and script traditions prefigure both the typed-event graph (Schank and Abelson, 1977; Lehnert, 1981) and the goal-directed planner / generator pipeline (Meehan, 1977; y Pérez and Sharples, 2001); the unsupervised narrative event chains of Chambers and Jurafsky (2008) are a direct precursor to the typed `causal_topology` maintained at runtime.

Cognitive narratology (Herman, 2002; Fludernik, 1996; Zwaan and Radvansky, 1998; Gergig, 1993) and constructionist comprehension (Trabasso and van den Broek, 1985; Graesser

et al., 1994; Kintsch and van Dijk, 1978). The five fields of the situation-model (time, space, protagonist, causation, intention) are nearly the field set on `EventNode`; that readers materialise causal links explicitly during comprehension is the empirical justification for materialising them explicitly in storage.

Theory of mind (Premack and Woodruff, 1978; Wimmer and Perner, 1983; Baron-Cohen et al., 1985; Zunshine, 2006; Baker et al., 2009) and its modern LLM benchmarks (Kim et al., 2023; Gu et al., 2026; Cross et al., 2024). `Belief{value, confidence, inertia, established_at_fabula, acquired_via_event_id, acquired_via_channel_id}` is a discrete approximation of recursive ToM with provenance. The benchmarks above motivate the auditor’s separate behaviour-given-belief consistency check.

Sternberg’s curiosity / suspense / surprise triad (Sternberg, 1978, 1992; Brewer and Lichtenstein, 1982) and its computational realisations (Cheong and Young, 2015; Bae and Young, 2008; Wilmot and Keller, 2020, 2021; Wilmot, 2022). The four structural scorers (MYS, IRO, SUSP, SUR) extend the triad with dramatic irony (Booth, 1974; Gergig, 1993). Suspense uses the hope/threat ledger of Wilmot and Keller (2020) but combines the two sides as a *balance* $\times$ *stakes* product, peaking under genuine outcome uncertainty rather than under one-sided dominance; surprise follows Itti and Baldi (2009) with a geometric prior update over a leave-one-out corpus marginal; emotional arc analytics owe to Reagan et al. (2016).

Information-theoretic and epistemic substrate (Shannon, 1948; Fagin et al., 1995; Alchourrón et al., 1985). `Channel` carries `intelligibility` $\in [0, 1]$ per participant (generalising the legacy encryption boolean) and an `eavesdropped_by` relation derived from the `intelligibility_threshold`; belief revision uses an AGM-style `beliefs_added` / `beliefs_invalidated` delta.

Causal inference (Pearl, 1995, 2009; Pearl and Mackenzie, 2018; Spirtes et al., 2000; Bareinboim et al., 2022) and actual causality (Halpern and Pearl, 2005; Halpern, 2016; Woodward, 2003). Pearl’s three rungs — rung 1

Correlation (observation), **rung 2 Interventions** (`do()`), and **rung 3 Counterfactuals** (abduction + `do`) — map directly to the query taxonomy used here (`ObservationQuery`, `InterventionQuery`, `CounterfactualQuery`); mechanism and `causal_force` on edges are motivated by Halpern’s “actual cause”; ego-graph slicing is a heuristic Markov-blanket cut (Verma and Pearl, 1988). Recent benchmarking of LLMs as causal reasoners (Kiciman et al., 2024) supports the hybrid stance taken here: LLMs propose edges (`extract_graph.py`); typed code identifies and propagates them (`causal_physics.py`).

Plan-and-render era of neural story generation (Riedl and Young, 2010; Jhala and Young, 2010; Martin et al., 2018; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Rashkin et al., 2020; Yang et al., 2022, 2023) and its benchmarks (Mostafazadeh et al., 2016; Fan et al., 2018; Akoury et al., 2020; Tian et al., 2024; Xu et al., 2025). The `CreativeBrief` (Step 9) plus channel-aware draft (Step 10) is a typed, causal-graph-conditioned variant of the same plan-then-render pattern. The brief enumerates per-effect typed `ConstraintBlocks`; the auditor (Step 11) is the rescore / revise step (Goldfarb-Tarrant et al., 2020; Yang et al., 2022) specialised to narrative-causal rather than stylistic criteria, in the LLM-as-judge tradition (Zheng et al., 2023; Bai et al., 2022; Madaan et al., 2023; Gu et al., 2024).

Reading as simulation (Mar and Oatley, 2008; Oatley, 2016, 1995; Hogan, 2003). The reason a tight causal/affective model matters more than surface fluency: fiction is a cognitive simulator, and what is being simulated is a typed world.

Game engines, interactive narrative, and world models for AI (Mateas and Stern, 2003; Riedl and Bulitko, 2013; Ware and Young, 2014; Kreminski et al., 2020; Park et al., 2023; Ha and Schmidhuber, 2018; LeCun, 2022; Hao et al., 2023; Wang et al., 2024; Smelik et al., 2014; Summerville et al., 2018). Architecturally, Shadow-Loom is closer to a game engine than to a sequence model: a typed simulation step (causal physics) is wrapped by a director / drama-manager (narrative physics) and exposed to a stochastic policy (the LLM renderer) only through a constrained action interface, in the lineage of Mimesis / Façade-style interactive drama (Mateas and Stern, 2003; Riedl

and Bulitko, 2013) and narrative planners such as CPOCL/Glaive (Ware and Young, 2014). The generative-agent line of work (Park et al., 2023) and the “world model” line in deep RL (Ha and Schmidhuber, 2018; LeCun, 2022; Hao et al., 2023) make a similar wager from the opposite direction: that an explicit, manipulable latent world is what makes long-horizon coherent behaviour tractable. The shadow branches in Shadow-Loom play the role those world models play — a sandbox in which counterfactual rollouts can be evaluated before being committed — but with a typed, inspectable schema in place of an opaque latent state. The bundled example worlds and authoring tools relate Shadow-Loom to the broader procedural-content-generation tradition (Smelik et al., 2014; Summerville et al., 2018; Kreminski et al., 2020).

4 What is novel, what is borrowed

Borrowed. Every ingredient in Section 3 is borrowed: Pearl’s ladder (Pearl, 2009; Bareinboim et al., 2022); the AMWN + ctf-calculus framework (Correa and Bareinboim, 2025); actual-causality semantics (Halpern, 2016); the suspense and surprise formalisms of Wilmot and Keller (2020) and Itti and Baldi (2009); the LLM-as-judge audit pattern (Bai et al., 2022; Madaan et al., 2023; Gu et al., 2024); the plan-and-render line in neural story generation (Riedl and Young, 2010; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Yang et al., 2023, 2022; Rashkin et al., 2020); the typed-event / story-grammar tradition (Schank and Abelson, 1977; Lehnert, 1981; Mani, 2012; Elson, 2012); and the narratological foundations (Shklovsky, 1965; Genette, 1980; Sternberg, 1992; Ryan, 1991; Herman, 2002; Zwaan and Radvansky, 1998).

Novel as a combination. I am not aware of another open-source pipeline that, taken together: (a) runs all three Pearl rungs over a typed narrative graph; (b) tags every node and edge with an AMWN `world_id` (Correa and Bareinboim, 2025) so that counterfactuals are first-class persisted objects; (c) couples that machinery to closed-form scorers for the four classical reader-states of narrative cognition; (d) uses the resulting score as a loss function over candidate interventions, with the LLM constrained to render the winner; and (e) closes the loop by re-extracting the rendered prose and comparing it against the brief. Each ingredient is borrowed. The composition is the modest contribution: a typed two-engine substrate

underneath a constrained renderer. Architecturally the closest analogue is something like a game engine for narrative: a typed simulation step gated by a drama-manager-like scorer (Mateas and Stern, 2003; Riedl and Bulitko, 2013; Ware and Young, 2014), driving a constrained stochastic actor in the spirit of recent generative-agent and world-model work (Park et al., 2023; Ha and Schmidhuber, 2018; Hao et al., 2023), but with the latent state replaced by an explicit causal-graphical schema.

Specific small constructions. The “Impact > Inertia” propagation rule (Eq. 1) and the geometric prior update inside the surprise scorer (Eq. 8) are introduced here as deliberately simple choices that keep both physics monotonic in useful ways (App. A.5). The two-axis fabula/syuzhet sampling of the affective scorers (App. A.5) is, to my knowledge, also new.

5 Rationale for the design choices

Why a typed graph rather than free text + RAG?

Recent stress tests show that LLMs can answer “what does X believe?” but fail to behave consistently with that belief in downstream action prediction (Gu et al., 2026; Kim et al., 2023). Belief and causal structure that are recoverable from prose are not, on present evidence, reliably *operated on* inside the prose-only loop. A typed graph makes belief, channel intelligibility, and causal mechanism inspectable and intervenable.

Why both a causal and a narrative physics?

The causal physics decides *what is allowed to happen* given the graph; the narrative physics decides *which of the allowed things would be most readable*. Decoupling licensing from desirability lets the system score and rank candidate interventions before any text is generated, and lets the same engine serve both “what-if” analytical queries (rung 2/3) and high-level affective directives (“maximise dramatic irony for character X”).

Why constrain the LLM rather than fine-tune one?

The renderer’s job is reduced to dialogue, description, and pacing within a mathematical envelope. Whatever the model’s training distribution, the brief enumerates the causal edges, beliefs to preserve, channels to keep hidden, and trait shifts to honour. Errors that survive into prose are then catchable by the audit pass because they correspond to specific, testable claims — the “compile and test” analogue from Motivation 2.

Why AMWN tagging at the persistence layer? Treating a counterfactual as a sibling `VersionRow` with `world_id="shadow"` makes “what-if” branches first-class, diffable, promotable, and auditable; it is the persistence-layer analogue of the AMWN graphical construction (Correa and Bareinboim, 2025). A user can compare a shadow against canon, keep both, or promote.

6 Why this might matter beyond fiction

NLP / reasoning. Narrative is, for these purposes, a reasonably tractable test bed for counterfactual and theory-of-mind reasoning, because the ground truth (what happens, who knows what) is finite and authored. A typed-graph + symbolic physics + constrained-renderer architecture is one concrete instantiation of the neuro-symbolic stance for controllable generation; the audit pass produces a falsifiable record of where the LLM departed from the brief. Recent ToM benchmarks (Gu et al., 2026; Kim et al., 2023; Cross et al., 2024) ask exactly the kind of behaviour-given-belief questions that an explicit `Belief` field with provenance is designed to support.

Computational social science and digital humanities.

Many socio-historical and humanistic questions are counterfactual in form (“what if policy X had not been adopted?”, “how would this novel read if the protagonist had spoken sooner?”). A graphical substrate that lets a researcher ingest a narrative source, fork a shadow branch under an explicit intervention, and inspect the propagated trait and relationship deltas is a small step toward auditable narrative simulation that can sit alongside established computational-social-science (Lazer et al., 2009) and digital-humanities tooling. I make no claim of empirical validity for any specific simulation; the contribution is the substrate, not its calibration.

Author tooling. The same machinery surfaces in a NiceGUI workspace and a Model-Context-Protocol server with around forty tools, exposing the graph, the affective scorers, and the version DAG to either human authors or LLM agents. The workspace is described component-by-component in App. D.

An invitation to other research communities.

The substrate is deliberately domain-agnostic at the graph level: entities, events, beliefs, channels, and

counterfactual branches are not specific to fiction. Computational social scientists (Lazer et al., 2009) routinely treat narrative sources — court records, parliamentary debates, oral histories, news streams — as data, and digital-humanities scholarship has long sought structured, sharable representations of literary works that can support both close and distant reading. Shadow-Loom may be read as one such representation: an open, typed, version-controlled graph with an explicit counterfactual sandbox and an audit trail back to the source text. Researchers in those communities, as well as in narratology and cognitive science, are warmly invited to fork the schema, swap in their own ingestion prompts and affective scorers, and report what they find. The pipeline ships with sixteen worked example worlds drawn from canonical literature and film, which may serve as a common reference set for cross-group experimentation.

7 Limitations and future evaluation

Shadow-Loom is an experimental research artefact rather than a benchmarked NLP system, and I want to be explicit about what that does and does not imply. No headline numbers are reported against an external dataset: the scores produced by the narrative physics are design choices grounded in narratological theory rather than validated reader-reaction models, and the causal physics borrows the AMWN naming and three-rule mental model of Correa and Bareinboim (2025) without implementing their identification algorithm literally — the graph is heavily typed for narrative use rather than for rigorous structural causal model identification. The ingestion pipeline depends on LLM extraction, and although a programmatic validation and auto-repair loop catches the more obvious failure modes, extraction errors can still propagate. The current unit-test suite covers the symbolic layers but does not amount to external evaluation.

Formal evaluation is therefore deferred to future work, along several complementary axes: (i) human-judgement studies of the affective scorers against reader annotations, in the spirit of Wilmot and Keller (2020); (ii) targeted ablations on bundled fixtures to quantify the contribution of the causal physics, the AMWN sandbox, and the audit loop to downstream coherence; and (iii) comparison against end-to-end LLM baselines on counterfactual and theory-of-mind benchmarks (Kim et al., 2023; Gu et al., 2026; Cross et al., 2024; Xu et al.,

2025). I share the present design as a substrate for further research — by myself and by others — not as a finished product.

Code, licence, and reproducibility

The code is released open source under AGPL-3.0-or-later, with a parallel commercial licence; the authoritative LICENSE file lives in the project repository. The pipeline, schema, MCP surface, auditor, and the affective scorers described here all sit in the `shadow_loom/` package. Worked examples on bundled plots (Macbeth, Death on the Nile, Reservoir Dogs, and others) ship in `example_worlds/` and `sample_plots/`.

References

- Nader Akoury, Shufan Wang, Josh Whiting, Stephen Hood, Nanyun Peng, and Mohit Iyyer. 2020. STORIUM: A dataset and evaluation platform for machine-in-the-loop story generation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Carlos E. Alchourrón, Peter Gärdenfors, and David Makinson. 1985. On the logic of theory change: Partial meet contraction and revision functions. *Journal of Symbolic Logic*, 50(2):510–530.
- Byung-Chull Bae and R. Michael Young. 2008. A use of flashback and foreshadowing for surprise arousal in narrative using a plan-based approach. In *Proceedings of the 1st International Conference on Interactive Digital Storytelling (ICIDS)*, volume 5334 of *LNCS*, pages 156–167.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, and 1 others. 2022. Constitutional AI: Harmlessness from AI feedback. ArXiv:2212.08073.
- Chris L. Baker, Rebecca Saxe, and Joshua B. Tenenbaum. 2009. [Action understanding as inverse planning](#). *Cognition*, 113(3):329–349.
- Mieke Bal. 2009. *Narratology: Introduction to the Theory of Narrative*, 3rd edition. University of Toronto Press.
- Alexander Balke and Judea Pearl. 1994. Counterfactual probabilities: Computational methods, bounds and applications. In *Proceedings of the 10th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 46–54.
- Elias Bareinboim, Juan D. Correa, Duligur Ibeling, and Thomas Icard. 2022. [On Pearl’s hierarchy and the foundations of causal inference](#). In *Probabilistic and Causal Inference: The Works of Judea Pearl*, pages 507–556. ACM Books.

- Simon Baron-Cohen, Alan M. Leslie, and Uta Frith. 1985. Does the autistic child have a 'theory of mind'? *Cognition*, 21(1):37–46.
- Wayne C. Booth. 1974. *A Rhetoric of Irony*. University of Chicago Press.
- David Bordwell. 1985. *Narration in the Fiction Film*. University of Wisconsin Press.
- Claude Bremond. 1973. *Logique du récit*. Seuil.
- William F. Brewer and Edward H. Lichtenstein. 1982. Stories are to entertain: A structural-affect theory of stories. *Journal of Pragmatics*, 6(5–6):473–486.
- Selmer Bringsjord and David Ferrucci. 2000. *Artificial Intelligence and Literary Creativity: Inside the Mind of BRUTUS, a Storytelling Machine*. Lawrence Erlbaum.
- Nathanael Chambers and Dan Jurafsky. 2008. Unsupervised learning of narrative event chains. In *Proceedings of the 46th Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Yun-Gyung Cheong and R. Michael Young. 2015. Suspenser: A story generation system for suspense. *IEEE Transactions on Computational Intelligence and AI in Games*, 7(1):39–52.
- Juan D. Correa and Elias Bareinboim. 2025. [Counterfactual graphical models: Constraints and inference](#). In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. Spotlight. OpenReview ID: Z1qZoHa6ql.
- Juan D. Correa, Sanghack Lee, and Elias Bareinboim. 2021. Nested counterfactual identification from arbitrary surrogate experiments. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2107.03190.
- Logan Cross, Violet Xiang, Agam Bhatia, Daniel L. K. Yamins, and Nick Haber. 2024. Hypothetical minds: Scaffolding theory of mind for multi-agent tasks with large language models. ArXiv:2407.07086.
- Lubomír Doležel. 1998. *Heterocosmica: Fiction and Possible Worlds*. Johns Hopkins University Press.
- David K. Elson. 2012. *Modeling Narrative Discourse*. Ph.D. thesis, Columbia University.
- Ronald Fagin, Joseph Y. Halpern, Yoram Moses, and Moshe Y. Vardi. 1995. *Reasoning About Knowledge*. MIT Press.
- Angela Fan, Mike Lewis, and Yann Dauphin. 2018. Hierarchical neural story generation. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Monika Fludernik. 1996. *Towards a 'Natural' Narratology*. Routledge.
- Gérard Genette. 1980. *Narrative Discourse: An Essay in Method*. Cornell University Press.
- Richard J. Gerrig. 1993. *Experiencing Narrative Worlds: On the Psychological Activities of Reading*. Yale University Press.
- Seraphina Goldfarb-Tarrant, Tuhin Chakrabarty, Ralph Weischedel, and Nanyun Peng. 2020. Content planning for neural story generation with Aristotelian rescoring. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4319–4338.
- Arthur C. Graesser, Murray Singer, and Tom Trabasso. 1994. Constructing inferences during narrative text comprehension. *Psychological Review*, 101(3):371–395.
- Algirdas Julien Greimas. 1983. *Structural Semantics: An Attempt at a Method*. University of Nebraska Press. Original work *Sémantique structurale* published 1966.
- Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, Yinghan Shen, Shengjie Ma, Honghao Liu, Saizhuo Wang, Kun Zhang, Yuanzhuo Wang, Wen Gao, Lionel Ni, and Jian Guo. 2024. A survey on LLM-as-a-judge. ArXiv:2411.15594.
- Yuling Gu, Oyvind Tafjord, Hyunwoo Kim, Jared Moore, Ronan Le Bras, Peter Clark, and Yejin Choi. 2026. SimpleToM: Exposing the gap between explicit tom inference and implicit tom application in LLMs. In *Proceedings of the International Conference on Learning Representations (ICLR)*. ArXiv:2410.13648; ICLR 2026.
- David Ha and Jürgen Schmidhuber. 2018. Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems 31 (NeurIPS)*.
- Joseph Y. Halpern. 2016. *Actual Causality*. MIT Press.
- Joseph Y. Halpern and Judea Pearl. 2005. [Causes and explanations: A structural-model approach. Part I: Causes](#). *British Journal for the Philosophy of Science*, 56(4):843–887.
- Shibo Hao, Yi Gu, Haodi Ma, Joshua Jiahua Hong, Zhen Wang, Daisy Zhe Wang, and Zhiting Hu. 2023. Reasoning with language model is planning with world model. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- David Herman. 2002. *Story Logic: Problems and Possibilities of Narrative*. University of Nebraska Press.
- Patrick Colm Hogan. 2003. *The Mind and Its Stories: Narrative Universals and Human Emotion*. Cambridge University Press.

- Laurent Itti and Pierre Baldi. 2009. [Bayesian surprise attracts human attention](#). *Vision Research*, 49(10):1295–1306.
- Arnav Jhala and R. Michael Young. 2010. Cinematic visual discourse: Representation, generation, and evaluation. *IEEE Transactions on Computational Intelligence and AI in Games*, 2(2):69–81.
- Emre Kıcıman, Robert Ness, Amit Sharma, and Chenhao Tan. 2024. Causal reasoning and large language models: Opening a new frontier for causality. *Transactions on Machine Learning Research*. ArXiv:2305.00050.
- Hyunwoo Kim, Melanie Sclar, Xuhui Zhou, Ronan Le Bras, Gunhee Kim, Yejin Choi, and Maarten Sap. 2023. FANToM: A benchmark for stress-testing machine theory of mind in interactions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics. ArXiv:2310.15421.
- Walter Kintsch and Teun A. van Dijk. 1978. Toward a model of text comprehension and production. *Psychological Review*, 85(5):363–394.
- Max Kreminski, Melanie Dickinson, Michael Mateas, and Noah Wardrip-Fruin. 2020. [Why are we like this?: The ai architecture of a co-creative storytelling game](#). In *Proceedings of the 15th International Conference on the Foundations of Digital Games (FDG)*.
- David Lazer, Alex Pentland, Lada Adamic, Sinan Aral, Albert-László Barabási, Devon Brewer, Nicholas Christakis, Noshir Contractor, James Fowler, Myron Gutmann, Tony Jebara, Gary King, Michael Macy, Deb Roy, and Marshall Van Alstyne. 2009. [Computational social science](#). *Science*, 323(5915):721–723.
- Yann LeCun. 2022. A path towards autonomous machine intelligence. Open Review preprint. Version 0.9.2.
- Wendy G. Lehnert. 1981. Plot units and narrative summarization. *Cognitive Science*, 5(4):293–331.
- David Lewis. 1973. [Causation](#). *Journal of Philosophy*, 70(17):556–567.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Jean M. Mandler and Nancy S. Johnson. 1977. Remembrance of things parsed: Story structure and recall. *Cognitive Psychology*, 9(1):111–151.
- Inderjeet Mani. 2012. *Computational Modeling of Narrative*. Morgan and Claypool.
- Raymond A. Mar and Keith Oatley. 2008. The function of fiction is the abstraction and simulation of social experience. *Perspectives on Psychological Science*, 3(3):173–192.
- Lara J. Martin, Prithviraj Ammanabrolu, Xinyu Wang, William Hancock, Shruti Singh, Brent Harrison, and Mark O. Riedl. 2018. Event representations for automated story generation with deep neural nets. In *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*.
- Michael Mateas and Andrew Stern. 2003. Façade: An experiment in building a fully-realized interactive drama. In *Game Developers Conference, Game Design Track*, San Jose, CA, USA.
- James R. Meehan. 1977. TALE-SPIN, an interactive program that writes stories. In *Proceedings of the 5th International Joint Conference on Artificial Intelligence (IJCAI)*.
- Nasrin Mostafazadeh, Nathanael Chambers, Xiaodong He, Devi Parikh, Dhruv Batra, Lucy Vanderwende, Pushmeet Kohli, and James Allen. 2016. A corpus and cloze evaluation for deeper understanding of commonsense stories. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*.
- Keith Oatley. 1995. A taxonomy of the emotions of literary response and a theory of identification in fictional narrative. *Poetics*, 23:53–74.
- Keith Oatley. 2016. Fiction: Simulation of social worlds. *Trends in Cognitive Sciences*, 20(8):618–628.
- Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. [Generative agents: Interactive simulators of human behavior](#). In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*.
- Thomas Pavel. 1986. *Fictional Worlds*. Harvard University Press.
- Judea Pearl. 1995. [Causal diagrams for empirical research](#). *Biometrika*, 82(4):669–688.
- Judea Pearl. 2009. [Causality: Models, Reasoning, and Inference](#), 2nd edition. Cambridge University Press.
- Judea Pearl and Dana Mackenzie. 2018. *The Book of Why: The New Science of Cause and Effect*. Basic Books.
- David Premack and Guy Woodruff. 1978. [Do the chimpanzees have a theory of mind?](#) *Behavioral and Brain Sciences*, 1(4):515–526.
- Vladimir Propp. 1968. *Morphology of the Folktale*, 2nd english edition. University of Texas Press.

- Hannah Rashkin, Asli Celikyilmaz, Yejin Choi, and Jianfeng Gao. 2020. PlotMachines: Outline-conditioned generation with dynamic plot state tracking. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Andrew J. Reagan, Lewis Mitchell, Dilan Kiley, Christopher M. Danforth, and Peter Sheridan Dodds. 2016. The emotional arcs of stories are dominated by six basic shapes. *EPJ Data Science*, 5:Article 31.
- Mark O. Riedl and Vadim Bulitko. 2013. Interactive narrative: An intelligent systems approach. *AI Magazine*, 34(1):67–77.
- Mark O. Riedl and R. Michael Young. 2010. Narrative planning: Balancing plot and character. *Journal of Artificial Intelligence Research*, 39:217–268.
- David E. Rumelhart. 1975. Notes on a schema for stories. In Daniel G. Bobrow and Allan M. Collins, editors, *Representation and Understanding: Studies in Cognitive Science*, pages 211–236. Academic Press.
- Marie-Laure Ryan. 1991. *Possible Worlds, Artificial Intelligence, and Narrative Theory*. Indiana University Press.
- Marie-Laure Ryan. 2001. *Narrative as Virtual Reality: Immersion and Interactivity in Literature and Electronic Media*. Johns Hopkins University Press.
- Roger C. Schank and Robert P. Abelson. 1977. *Scripts, Plans, Goals and Understanding: An Inquiry into Human Knowledge Structures*. Lawrence Erlbaum.
- Claude E. Shannon. 1948. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656.
- Viktor Shklovsky. 1965. Art as technique. In Lee T. Lemon and Marion J. Reis, editors, *Russian Formalist Criticism: Four Essays*, pages 3–24. University of Nebraska Press, Lincoln, NE. Original work published 1917.
- Ilya Shpitser and Judea Pearl. 2008. Complete identification methods for the causal hierarchy. *Journal of Machine Learning Research*, 9(64):1941–1979.
- Ruben M. Smelik, Tim Tutenel, Rafael Bidarra, and Bedrich Benes. 2014. A survey on procedural modelling for virtual worlds. *Computer Graphics Forum*, 33(6):31–50.
- Peter Spirtes, Clark Glymour, and Richard Scheines. 2000. *Causation, Prediction, and Search*, 2nd edition. MIT Press.
- Meir Sternberg. 1978. *Expositional Modes and Temporal Ordering in Fiction*. Johns Hopkins University Press.
- Meir Sternberg. 1992. Telling in time (II): Chronology, teleology, narrativity. *Poetics Today*, 13(3):463–541.
- Adam Summerville, Sam Snodgrass, Matthew Guzdial, Christoffer Holmgård, Amy K. Hoover, Aaron Isaksen, Andy Nealen, and Julian Togelius. 2018. Procedural content generation via machine learning (PCGML). *IEEE Transactions on Games*, 10(3):257–270.
- Yufei Tian, Tenghao Huang, Miri Liu, Derek Jiang, Alexander Spangher, Muhao Chen, Jonathan May, and Nanyun Peng. 2024. Are large language models capable of generating human-level narratives? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 17659–17681. Association for Computational Linguistics. Outstanding Paper Award.
- Boris Tomashevsky. 1965. Thematics. In Lee T. Lemon and Marion J. Reis, editors, *Russian Formalist Criticism: Four Essays*, pages 61–95. University of Nebraska Press.
- Tom Trabasso and Paul van den Broek. 1985. Causal thinking and the representation of narrative events. *Journal of Memory and Language*, 24(5):612–630.
- Thomas Verma and Judea Pearl. 1988. Causal networks: Semantics and expressiveness. In *Proceedings of the 4th Workshop on Uncertainty in Artificial Intelligence (UAI)*, pages 69–78.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2024. Voyager: An open-ended embodied agent with large language models. In *Transactions on Machine Learning Research*. ArXiv:2305.16291.
- Stephen G. Ware and R. Michael Young. 2014. Glaive: A state-space narrative planner supporting intentionality and conflict. In *Proceedings of the Tenth AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, pages 80–86.
- David Wilmot. 2022. *Great Expectations: Unsupervised Inference of Suspense, Surprise and Salience in Storytelling*. Ph.D. thesis, University of Edinburgh. ArXiv:2206.09708.
- David Wilmot and Frank Keller. 2020. Modelling suspense in short stories as uncertainty reduction over neural representation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1763–1788. Association for Computational Linguistics.
- David Wilmot and Frank Keller. 2021. Memory and knowledge augmented language models for inferring salience in long-form stories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 851–865.
- Heinz Wimmer and Josef Perner. 1983. Beliefs about beliefs: Representation and constraining function of wrong beliefs in young children’s understanding of deception. *Cognition*, 13(1):103–128.

James Woodward. 2003. *Making Things Happen: A Theory of Causal Explanation*. Oxford University Press.

Weijia Xu, Nebojsa Jojic, Sudha Rao, Chris Brockett, and Bill Dolan. 2025. [Echoes in AI: Quantifying lack of plot diversity in LLM outputs](#). *Proceedings of the National Academy of Sciences*, 122(35):e2504966122. ArXiv:2501.00273.

Rafael Pérez y Pérez and Mike Sharples. 2001. MEXICA: A computer model of a cognitive account of creative writing. *Journal of Experimental & Theoretical Artificial Intelligence*, 13(2):119–139.

Kevin Yang, Dan Klein, Nanyun Peng, and Yuandong Tian. 2023. DOC: Improving long story coherence with detailed outline control. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*. ArXiv:2212.10077.

Kevin Yang, Yuandong Tian, Nanyun Peng, and Dan Klein. 2022. Re3: Generating longer stories with recursive reprompting and revision. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Lili Yao, Nanyun Peng, Ralph Weischedel, Kevin Knight, Dongyan Zhao, and Rui Yan. 2019. Plan-and-write: Towards better automatic storytelling. In *Proceedings of the 33rd AAAI Conference on Artificial Intelligence*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

Lisa Zunshine. 2006. *Why We Read Fiction: Theory of Mind and the Novel*. Ohio State University Press.

Rolf A. Zwaan and Gabriel A. Radvansky. 1998. [Situation models in language comprehension and memory](#). *Psychological Bulletin*, 123(2):162–185.

A Definitions and equations

This appendix gives the formal definitions of the data model and the equations of each pipeline stage. Section references in the main paper point here.

A.1 World model schema (WorldStateV1)

A world state is a tuple

$$\mathcal{W} = (\mathcal{N}, \mathcal{E}, \mathcal{C}, \mathcal{T}, \tau)$$

with nodes $\mathcal{N}$ (entities ENT_●, events EVT_●, locations LOC_●, objects OBJ_●, channels CHN_●, world traits WORLD_●), edges $\mathcal{E}$ (causal, relationship, spatial), channels $\mathcal{C}$ with per-participant intelligibility $\iota \in [0, 1]$, world traits $\mathcal{T}$, and a branch tag $\tau \in \{\text{factual}, \text{shadow}\}$ inherited by every node and edge.

Each event $e \in \mathcal{N}_{\text{EVT}}$ carries two integer indices: chronological $f(e) \in \mathbb{Z}$ (*fabula time*) and presentation $s(e) \in \mathbb{Z}$ (*syuzhet index*), with s contiguous and unique. Trait values, beliefs, and ambient state are carried by typed vectors $(v, \iota_{\text{inertia}}, \sigma_{\text{evidence}})$ representing value, inertia, and evidence strength. Beliefs carry provenance (*event_id*, *channel_id*) and the time they became established. A pure function $\text{RECONSTRUCT}(e_i, t)$ replays sparse *state_timeline* deltas to give the entity’s effective state at any fabula time t .

A.2 Ingestion

Given prose P , ingestion produces a $\mathcal{W}$ via:

1. Five extraction agents over P build a `GlobalRegister` of `WORLD_`, `ENT_`, `LOC_`, `OBJ_` nodes plus a fuzzy alias table.
2. For each chunk, a Socratic Who/What/Where/When/Why/How scaffold dispatches three concurrent specialist agents: *physics* (events, causal/spatial edges), *social* (relationship edges, channels), *consequences* (authoritative entity-state deltas).
3. Output is normalised, fabula times are remapped to a uniform $\Delta t = 100$ spacing, and a programmatic validator + LLM correction loop enforces ID closure and type constraints.

A.3 Ego-graph slicing and AMWN sandbox

Given a query with focal entities F and time anchor t , the ego-graph extractor returns

$$\mathcal{G}_{F,t} = (\text{NEIGHBOURS}_k(F) \cap \text{RECONSTRUCT}(\cdot, t)),$$

the k -hop spatial / causal / social neighbourhood of F at fabula time t , with belief and trait values reconstructed at t so future information cannot leak into a counterfactual. The `AMWNInstantiator` mirrors $\mathcal{G}_{F,t}$ as a `NetworkX MultiDiGraph`, in the spirit of the AMWN construction (Correa and Bareinboim, 2025), with edge types `{located_in, owned_by, causal, relationship, connected_to, communicating_with, eavesdropped_by}`.

A.4 Causal physics

The causal physics is a *graphical world model of cause and effect*: a typed dynamics over the shared world graph that implements all three rungs of Pearl’s causal hierarchy (Pearl, 2009; Bareinboim et al., 2022): **rung 1 is Correlation** (observation: $P(Y \mid X)$ read off the ego-graph without intervention), **rung 2 is Interventions** ($\text{do}(X = x)$, severing incoming edges into X), and **rung 3 is Counterfactuals** (abduction over latent exogenous values, then a do -operation on the abducted world). Each rung corresponds to one query type in Section ??: `ObservationQuery`, `InterventionQuery`, and `CounterfactualQuery` respectively.

Given a sandbox $\mathcal{G}$ and an intervention specification:

Observation (rung 1 — Correlation): read $P(Y \mid X = x)$ directly off the ego-graph; no edge surgery.

Action (rung 2 — Interventions): $\text{do}(X = x)$ severs every incoming causal edge into X and writes $V_X \leftarrow x$.

Abduction (rung 3 — Counterfactuals): for evidence E , each entity trait blends 50% toward its observed factual value, weighted by edge σ_{evidence} and gated by a typed `MECHANISM_TRAIT_MAP` that says which mechanisms can touch which trait families. The do-operation of rung 2 is then re-applied on the abducted world.

Propagation: a topological sort over the causal sub-graph applies, for each downstream node u with parent p and edge weight w :

$$I(u) = (V_p - V_u) w, \quad \Delta_u = \begin{cases} I(u) - \text{sgn}(I(u)) \iota_u & \text{if } |I(u)| > \iota_u, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

A spatial affordance check additionally blocks cross-location influence unless an unbroken `connected_to` path exists. The result is a structured `CausalPhysicsResult{mutations, blocked, hidden_deltas, intervened_nodes}`.

A.5 Narrative physics (affective calculus)

The narrative physics is a *graphical world model of reader affect*: it operates on the same typed graph as the causal physics, but its dynamics scores reader-state functionals (mystery, dramatic irony, suspense, surprise) rather than propagating causal force.

Let A be the anchor pair (t_f, t_s) for fabula and syuzhet cursors, F the focal entities, and write *revealed* for any event e with $s(e) \leq t_s$. Define $\text{Anc}(e)$ as the causal ancestors of e in the graph and $B(F, t_f)$ as the union of belief sets of F reconstructed at t_f .

Mystery. For each revealed effect e ,

$$\text{Mys}(e) = \frac{|\{a \in \text{Anc}(e) : s(a) > t_s\}|}{|\text{Anc}(e)|}, \quad (2)$$

the fraction of e 's causes still hidden from the reader.

Dramatic irony. For each focal entity $c \in F$ let K_c be the set of events c knows at the anchor: events c participates in (as actor or target) whose fabula time falls at or before the syuzhet anchor's fabula frontier, plus events referenced in revealed utterances spoken by or addressed to c , plus events c holds a `Belief` about. Writing w_e for event e 's narrative intensity (default 1), the irony score is the per-character mean intensity-weighted mass of revealed events the character does not know about, normalised by the total event mass plus a saturation constant K :

$$\text{Iro} = \frac{1}{|F|} \sum_{c \in F} \frac{\sum_{e \in \text{rev}, e \notin K_c} w_e}{\sum_e w_e + K}, \quad K = 1. \quad (3)$$

The earlier ratio `#gaps/#revealed-edges` grew its numerator and denominator together and so plateaued at a story-specific asymptote by the third reveal — on *Reservoir Dogs* it even *decayed* from 0.25 to 0.06 as the protagonist became actor-of-record on more revealed edges. Normalising by the full event mass (a fixed denominator) lets the gap mass climb monotonically with reveals and *fall* when characters acquire knowledge later in the story — the dramatic-irony arc the formal definition is supposed to track.

Suspense (Wilmot and Keller, 2020). For each unrevealed event e with strongest incoming causal weight $p_e \in (0, 1]$, classify focal entities per-event: an entity acted upon without itself acting contributes to threat, an entity acting contributes to hope. Aggregating these per-entity contributions:

$$w_{\text{threat}} = \sum_{\substack{e \notin \text{rev} \\ x \in F \cap \text{target}(e) \setminus \text{actor}(e)}} p_e, \quad w_{\text{hope}} = \sum_{\substack{e \notin \text{rev} \\ x \in F \cap \text{actor}(e)}} p_e. \quad (4)$$

Let $T = w_{\text{threat}} + w_{\text{hope}}$. Define

$$\text{balance} = 1 - \frac{|w_{\text{threat}} - w_{\text{hope}}|}{T}, \quad \text{stakes} = \frac{T}{T + K}, \quad (5)$$

so $\text{balance} \in [0, 1]$ peaks at genuine outcome uncertainty ($w_{\text{threat}} = w_{\text{hope}}$) and decays to 0 under one-sided dominance, while $\text{stakes} \in [0, 1)$ saturates so balanced fragments do not pin the gauge. Then

$$\text{Susp} = \text{clip}_{[0,1]}(\text{balance} \cdot \text{stakes}), \quad (6)$$

returning 0 when $w_{\text{hope}} = 0$ (the suspense–despair boundary) or when $w_{\text{threat}} = 0$ (safety). The default $K = 2$ is calibrated so two strong unrevealed events on each side already register as fully high-stakes; this replaces an earlier asymmetric $\max(0, (w_{\text{threat}} - w_{\text{hope}})/T)$ form that collapsed to 0 on every fixture in which the protagonist authors most of their own forward events.

Surprise. Let $\text{actual}_T(x)$ denote entity x ’s trait T value at the world’s terminal fabula time, resolved via $\text{RECONSTRUCT}(x, t_{\text{max}})$ so authored `state_timeline` arcs (e.g. Macbeth’s ambition rising $0.7 \rightarrow 0.85$) are honoured rather than read off the baseline. For each focal entity $x \in F$, the prior q starts at the *leave-one-out* corpus marginal

$$q_0(T, x) = \frac{1}{|\mathcal{N}_{\text{ENT}} \setminus \{x\}|} \sum_{y \in \mathcal{N}_{\text{ENT}} \setminus \{x\}} \text{actual}_T(y), \quad (7)$$

defaulting to 0.5 when fewer than two other entities carry T . Leave-one-out prevents the focal entity from biasing its own prior — with the small casts typical of the example fixtures the inclusive marginal would otherwise pull q toward p and systematically depress KL.

For each revealed cause ($a \rightarrow x$) with weight w the prior is updated by

$$q += w (\text{actual}_T(x) - q), \quad (8)$$

a geometric pull that monotonically converges on the truth (an additive $w(p - q_0)$ form would summit past p once $\sum w > 1$, producing non-monotonic surprise). Letting $p = \text{actual}_T(x)$,

$$D_{\text{KL}}(p \parallel q) = p \log \frac{p}{q} + (1 - p) \log \frac{1-p}{1-q}, \quad (9)$$

and the entity-level surprise is the mean over traits of the soft-saturated per-trait KL,

$$\text{Sur} = \frac{1}{|\mathcal{T}|} \sum_{T \in \mathcal{T}} (1 - e^{-D_{\text{KL}}(p \parallel q)}), \quad (10)$$

which maps perceptually meaningful KLs in the 0.2–1.5 band to the 0.18–0.78 region of the gauge. The earlier $\overline{\text{KL}} / \log(1/\varepsilon)$ form divided by the *theoretical* binary-KL maximum ($\log(1/\varepsilon) \approx 4.6$ at $\varepsilon = 0.01$), squashing the entire perceptual signal into the bottom 4% of the gauge — every example-world plot read as flat ≤ 0.10 even when canonical surprise traits (Macbeth’s despair, Macduff’s grief, Lady Macbeth’s guilt) carried per-trait KLs of 0.27–0.50. With no syuzhet anchor the reader is assumed omniscient and $\text{Sur} = 0$.

Six emotion scorers (*grief, rage, joy, regret, love, fear*) score *closeness* to a per-effect trait target. A shared `_EFFECT_TRAITS` table maps each effect to a pair (P_e, I_e) of *positive* and *inverse* trait indicators, calibrated against the actual trait vocabulary of the example-world corpus. Letting $v(x, T)$ denote the current value of trait T on entity x (resolved via `RECONSTRUCT`), the per-entity score is

$$\text{Emo}_e(x) = \frac{\sum_{T \in P_e \cap \tau(x)} v(x, T) + \sum_{T \in I_e \cap \tau(x)} (1 - v(x, T))}{|(P_e \cup I_e) \cap \tau(x)|}, \quad (11)$$

where $\tau(x)$ are the trait names x carries. Both lists drive scoring symmetrically: an earlier filter applied P_e only, leaving I_e as dead code (a brave character routed through `fear` got no fear-reducing credit because `courage` never entered the average). When the intersection is empty — a deliberately traitless cast for that effect — the score falls back to the worst-case +1 rather than silently scoring at the midpoint. The combined affective score requested by a `DirectiveQuery` is a weighted sum over the active functionals.

Two-axis sampling. The same scorers are evaluated along the syuzhet axis (advancing t_s only reveals more, so mystery, irony, surprise are non-increasing) and the fabula axis (snapshotting the world at t_f , where surprise can spike at intrinsically surprising story moments).

A.6 Generation

`DirectiveAssembler.evaluate_candidate_events()` forks the sandbox per candidate intervention, runs the causal physics, prunes candidates that violate inertia, affordance, or propagation constraints, ranks survivors by the affective score, and packages the winner as a `CreativeBrief` of typed `ConstraintBlocks` (per-effect): hidden-predecessor lists for mystery, “MUST NOT learn” guards for irony, threat/hope tables and hidden-channel lists for suspense, per-trait KL magnitudes for surprise, and per-trait shift constraints with headroom and inertia evidence for the emotion targets. Only the brief is shown to the renderer LLM, which produces dialogue, description, and pacing within the envelope.

A.7 Audit and feedback loop

The auditor (an LLM-as-judge in the sense of [Gu et al., 2024](#)) runs three passes over the rendered prose: (i) **causal** — reverse-engineers prose into causal claims and flags “miracle steps” (state changes with no licensing edge in the brief); (ii) **abduction** — runs counterfactual probes against the prose to check that implicit events hold; (iii) **affective** — measures the realised epistemic gap in the prose against the target. If the structured loss is non-zero, generation is re-run with the auditor’s feedback appended, up to `max_correction_retries`. On success the re-extracted prose is merged into a new `VersionRow`, with `world_id` set by the branch policy (`auto | mainline | shadow`); counterfactual queries default to a fresh shadow branch, preserving canon.

B Worked examples on real plot fixtures

The repository ships sixteen hand-authored `example_worlds/` fixtures paired with prose synopses in `sample_plots/`. A small subset is used below to illustrate each component end-to-end. Field values quoted below are verbatim from the fixtures; queries are the same ones run by the example walkthroughs in `docs/model-examples.md` and `docs/pipeline-by-example.md`.

B.1 World model — Macbeth

In `example_worlds/macbeth.py`, the location `LOC_HEATH` carries `ambient_state{supernatural: AmbientVector(value=0.9, volatility=0.2, evidence_strength="strong"), concealment: 0.7}`. The entity `ENT_MACBETH` is a bundle of typed `TraitVectors`: ambition, courage (high inertia 0.7, hard to shake), guilt (low inertia 0.25, highly mutable), and so on; an initial `Belief(target=PROPHECY, confidence=0.4)` sits on his belief list with its own inertia. `OBJ_BLOODY_DAGGERS` declares `Affordance(action="kill", target_type="Entity")` — the precondition gate the engine checks before it lets `EVT_DUNCAN_MURDER` fire. Every `EventNode` carries both indices: the murder has a fabula time but the syuzhet index follows the staged scene. The fixture exercises all five `CausalEdge` modalities (`chain_reaction`, `mutation`, `mutation_social`, `affordance_gate`, `ambient_propagation`); the `ambient_propagation` edge is what lets `LOC_HEATH.supernatural=0.9` bias Macbeth’s uptake of the prophecy belief.

B.2 AMWN sandbox — Macbeth “what if Macbeth refuses?”

The query “Macbeth refuses to murder Duncan” is a rung-2 intervention. `AMWNInstantiator.create_sandbox(F={ENT_MACBETH}, t)` mirrors the ego-graph as a `MultiDiGraph`; a sibling `VersionRow` with `world_id="shadow"` holds the fork. `apply_do_operator({EVT_DUNCAN_MURDER: ∅})` severs the murder’s incoming causal edges; the downstream `chain_reaction` edges into `EVT_MALCOLM_FLEES`, `EVT_MACBETH_CROWNED`, `EVT_BANQUO_MURDERED` all lose activation; the `mutation` edges into

`ENT_MACBETH.guilt` and `paranoia` never fire. The factual mainline is untouched — the shadow row is browsable, diffable, and promotable from the UI.

B.3 Causal physics — Macbeth propagation under “Impact > Inertia”

After the do-operator, `propagate()` topologically sorts the causal sub-graph. For the original (factual) chain, the edge `EVT_BANQUO_GHOST` $\rightarrow$ `ENT_MACBETH.guilt` with weight $w = 0.7$ and source delta $V_p - V_u = 0.6$ produces $I = 0.42$. Because `guilt`’s post-shock inertia ι has been bumped to ~ 0.3 from its baseline 0.25 by the previous `state_timeline` entry, the gate $|I| > \iota$ passes and $\Delta = 0.42 - 0.3 = 0.12$ is written. A symmetrical attempt to shock `courage` (initial inertia 0.7) at $w = 0.5$, $\Delta_{\text{src}} = 0.4$ gives $I = 0.20 < 0.7$ and is *blocked* — exactly the play’s psychology, in which his nerve outlasts his conscience. Spatial affordance gating additionally blocks cross-location shocks unless a `connected_to` path exists.

B.4 Narrative physics — Death on the Nile, Reservoir Dogs, Romeo and Juliet, Macbeth

The four structural scorers each have a canonical fixture.

Mystery (Macbeth Act II). Eq. 2 on the revealed effect `EVT_DUNCAN_FOUND_DEAD`: of its causal ancestors, `EVT_LADY_PERSUADES` and `EVT_DAGGERS_PLACED` are still hidden at the audience’s `syuzhet_anchor` early in the act; the score is $2/3 \approx 0.67$, falling to 0 once Act V resolves the provenance.

Dramatic irony (Death on the Nile). The reader knows Jacqueline and Simon are conspirators (`EVT_PLAN_HATCHED` revealed at low `syuzhet`); Poirot and the suspect pool do not. With $F = \{\text{ENT_PENNINGTON}, \text{ENT_VAN_SCHUYLER}, \text{ENT_ALLERTON}\}$, the conspiracy and its downstream revealed effects sit in the $\sum_{e \in \text{rev}, e \notin K_c} w_e$ numerator of Eq. 3 for every $c \in F$. The gauge climbs through the cruise scenes as Poirot’s reveals accumulate without reaching the suspect pool, and falls only at the denouement when Poirot briefs the cast and the events enter every K_c .

Suspense (Romeo and Juliet, tomb scene). At the tomb $F = \{\text{ENT_ROMEO}\}$, unrevealed events with Romeo as non-acting target (Friar’s letter intercepted, Juliet’s draught about to wear off) sum to w_{threat} ; events Romeo himself authors and that have not yet fired (drink poison, kill Paris) sum to w_{hope} . Threat dominates. As Romeo drinks, $w_{\text{hope}} \rightarrow 0$ and the score returns 0 — the Wilmot–Keller suspense–despair boundary is hit cleanly.

Surprise (Reservoir Dogs reveal). `EVT_ORANGE_RECRUITED` sits at low fabula time but high `syuzhet`; before the reveal the corpus-marginal prior on `ENT_ORANGE.role` starts at $q = 0.5$, and the few revealed edges into Orange give a small geometric pull keeping $q \approx 0.6$ “criminal”. After the reveal the actual value $p \approx 0.95$ “cop”. Eq. 8 gives a per-trait KL spike that is large but *licensed* — the engine had the cause present at low fabula time, only locked behind `syuzhet`, so re-extraction will not flag a miracle step.

Emotion targets (Gone Girl, Brief Encounter). The six emotion scorers (`grief`, `rage`, `joy`, `regret`, `love`, `fear`) resolve their effect through the shared `_EFFECT_TRAITS` table (Eq. 11). On *Gone Girl*, the directive “maximise grief in Nick” is scored against `grief`’s positive indicators (`despair`, `shame`, `longing`, `grief`, `anguish`, `mourning`) and inverse indicators (`hope`, `joy`, `contentment`, `satisfaction`) intersected with Nick’s actual trait set, with the available headroom against those targets determining whether the directive is even feasible at the chosen anchor. On *Brief Encounter* the same machinery realises `regret` as the gap between Laura’s current `self_actualisation` and the `regret`-positive trait values `guilt`, `remorse`, `despair` under a constant-supervision channel that never permits the choice to be made.

B.5 Directive assembly — Romeo and Juliet

The directive “raise dramatic irony to 0.85 in the tomb scene without giving Romeo correct information” triggers `DirectiveAssembler.evaluate_candidate_events()`. From unblocked

affordances and reachable spatial paths it enumerates candidates — *Friar’s letter intercepted*, *Balthasar arrives faster*, *Juliet wakes one minute earlier* — and forks an AMWN sandbox per candidate. The plausibility gate drops *Balthasar arrives faster* (the plague quarantine `SpatialEdge.is_locked` blocks the path); the remaining sandboxes are propagated, and the survivors are scored. *Friar’s letter intercepted* wins because it raises Eq. 2’s sister gap-count for irony without adding any utterance event into $B(\text{ENT_ROMEO}, t_f)$. The winner is wrapped in typed `ConstraintBlocks`: a `must_events` list, a `must_not_events` list (Romeo learning the truth), trait envelopes (Romeo’s despair envelope), and a `syuzhet-window` constraint. That bundle is the *only* thing handed to the renderer.

B.6 Generation — the brief as safety envelope

The renderer LLM is given the Romeo-and-Juliet brief as system prompt plus a small style context. It may write any prose about the tomb, the messenger, the dagger; it may not invent a causal edge, shift a trait outside its envelope, open a channel that does not exist, or place an utterance the brief does not licence. The output is a scene of dialogue, description, and pacing inside the mathematical envelope chosen by directive assembly — a typed, causal-graph-conditioned instance of the plan-and-render pattern (Riedl and Young, 2010; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Yang et al., 2023).

B.7 Audit and feedback loop — Gone Girl

Gone Girl is the system’s lying-narrator fixture: Amy’s diary is a sequence of `EventNodes` with `truth_value="false"` broadcast through `CHN_DIARY_PUBLIC`. After generation, `auditor.run_feedback_loop` runs three passes in parallel. The *causal audit* reverse-engineers the prose into (source, target, modality) claims and matches each against the brief; an LLM-emitted “Nick suddenly realises Amy is alive” with no licensing causal edge is flagged as a *miracle step*. The *abduction audit* runs counterfactual probes (“if the diary were truthful, would the prose still hold?”); a “yes” here means the prose has collapsed Amy’s two faces into one, and is rejected. The *affective audit* measures the realised dramatic-irony gap against the target. A *style-fidelity audit* compares the prose against the `NarrativeStyle` profile recovered at ingestion — word-count budget, prose density, register, POV, tense, and form class (`news_article`, `historical_account`, `thought_experiment`, etc.) — raising `style_mismatch` on drift. A *meta-narration audit* runs on counterfactual and abduction-driven scenes: it flags prose that comments on its own branch structure (*timeline*, *divergence*, *the alternative holds*, conditional/subjunctive framings used to *describe* rather than *narrate* the counterfactual) instead of rendering the alternate world as a lived past-tense scene, raising `meta_narration`. The aggregated `FeedbackLoopResult` exposes both an `engine_thresholds_passed` gate (deterministic: number of miracle steps, max trait drift) and an `llm_pass` gate (literary critique); either failing re-runs generation with the audit feedback appended, up to `max_correction_retries`. On convergence the scene is re-extracted into a new `VersionRow`; counterfactual branches land on `world_id="shadow"` and only become canon via explicit `promote_branch` (Correa and Bareinboim, 2025).

C End-to-end walkthrough on Macbeth

This appendix describes a single complete pass of the pipeline in plain prose. The goal is to make every intermediate object inspectable to a reader who has read the main text but has not read the source. The fixture used throughout is `example_worlds/macbeth.py`, paired with the synopsis in `sample_plots/macbeth.txt`; the query is the canonical counterfactual “what if Macbeth refuses to murder Duncan?”. The same loop runs identically on the other fifteen bundled fixtures.

C.1 Step 0: the source text and the user’s query

The user opens the workspace, drops the synopsis into the `STORY` tab, and clicks *Save & Re-ingest*. The synopsis is a five-act prose summary in the public-domain register: the witches on the heath, Duncan’s offer of the Cawdor title, Lady Macbeth’s persuasion, the murder in Inverness, the banquet, the apparitions, the slaughter at Macduff’s castle, the sleepwalking scene, Birnam Wood, and the final duel. After ingestion the user types a query into the `REASONING` tab: “*Counterfactual: Macbeth*

refuses to murder Duncan at Inverness. Show what changes for Macduff, Banquo, Malcolm, and the witches, and re-render the banquet scene.” The query is parsed by `query_parsing.parse_query` into a typed `CounterfactualQuery` with focal entities $F = \{\text{ENT_MACBETH}\}$, target $X = \text{EVT_DUNCAN_MURDER}$, intervention value $x = \emptyset$ (do-not-occur), and a re-render directive scoped to the banquet’s syuzhet window.

C.2 Step 1: ingestion in detail

Ingestion is the only stage where the LLM is allowed to author graph structure rather than prose. It is a fan-out / fan-in pattern over the chunked source.

Ontology pre-pass. Five extraction agents run over the full text in parallel: `ontology_locations` produces `LOC_HEATH`, `LOC_INVERNESS_CASTLE`, `LOC_FORRES_COURT`, and so on, each with an initial `ambient_state` dictionary; `ontology_objects` extracts `OBJ_BLOODY_DAGGERS`, `OBJ_LETTER`, and `OBJ_CROWN` with explicit Affordance lists; `ontology_entities` produces `ENT_MACBETH`, `ENT_LADY_MACBETH`, `ENT_DUNCAN`, and the rest, each as an Entity with a sparse initial `TraitVector` bundle, beliefs, and a first-appearance `location_id`; `ontology_world_traits` introduces the world-level named latents (`WORLD_AMBITION`, `WORLD_FATE`, `WORLD_GUILT`) that act as common causes; and `ontology_aliases` builds a fuzzy alias table (“Lord of Glamis” $\mapsto$ `ENT_MACBETH`). The result is a populated `GlobalRegister`.

Per-chunk specialist fan-out. The text is chunked into ~ 800 -token windows aligned on paragraph boundaries. Each chunk dispatches three concurrent specialist agents that share the `GlobalRegister` as system context: `physics_extraction` produces `EventNodes` with `event_type` drawn from `\{choice, outcome, revelation, utterance\}`, causal edges with `causality_type`, and spatial edges (e.g. a `connected_to` edge from `LOC_INVERNESS_CASTLE` to `LOC_FORRES_COURT`); `social_extraction` produces `RelationshipEdges` with per-axis `RelationshipMetrics` for affinity, fear, and power dynamic, and the Channels (the witches’ prophecy channel, the dinner-table channel at the banquet, the diary-style private channel of Lady Macbeth’s letter); and `consequences_extraction` writes the authoritative `state_timeline` entries on each entity, including the inertia bumps that record how shocking events make traits *harder* to dislodge afterwards (Macbeth’s `guilt` inertia rises from 0.25 to 0.4 after the murder). The three prompts share a *symmetric per-axis coverage rule*: whenever `social_extraction` emits a `RelationshipMetric` on any of the three axes, `physics_extraction` must emit at least one `mutation_social` edge with the matching `trait_target`; the validation pass enforces the parity. The rule prevents *static dyads* — relationships whose value is non-zero but never causally moved — from producing flat horizontals on the downstream conflict, danger, and power gauges. The bundled corpus (§B) honours this invariant on every observed axis on every dyad.

Validation and auto-repair. The merged graph passes through `_programmatic_validation`, which checks ID closure (every `source_id` resolves), type constraints (every `causality_type` is one of the five enums), basic acyclicity within causal edges of the same fabula slice, and “alive-actor” constraints (no event sourced from a dead character). Failures are passed back to a small repair LLM with the offending field and a structured error; the repair loop runs up to three iterations before raising. Fabula times are then remapped to a uniform $\Delta t = 100$ spacing so that downstream windows are stable across re-ingestion. The output is a single `WorldStateV1`, persisted as the root `VersionRow`.

C.3 Step 2: query routing and ego-graph slicing

The `CounterfactualQuery` is dispatched to the `rung-3` handler. The handler first consults the `NarrativePhysics` router, which selects an *ego-graph* around the focal entities at the query’s fabula anchor: Macbeth’s k -hop neighbourhood at $t \approx 6000$ (the Inverness night). The slice contains `ENT_MACBETH`, `ENT_LADY_MACBETH`, `ENT_DUNCAN`, the chamber attendants, `OBJ_BLOODY_DAGGERS`, `LOC_INVERNESS_CASTLE`, the relevant beliefs, and the immediate causal context (`EVT_LADY_PERSUADES`, the prophecy events, the upcoming `EVT_DUNCAN_MURDER`). Trait

values are reconstructed at t by replaying `state_timeline` deltas, which prevents post-murder trait shocks from leaking into the counterfactual past.

C.4 Step 3: AMWN sandbox and the do-operator

`AMWNInstantiator.create_sandbox(F={ENT_MACBETH}, t)` mirrors the ego-graph as a `NetworkX MultiDiGraph`, in the spirit of the Ancestral Multi-World Network construction of [Correa and Bareinboim \(2025\)](#). A sibling `VersionRow` with `world_id="shadow"` is opened to hold the fork. The do-operator `apply_do_operator({EVT_DUNCAN_MURDER:  $\emptyset$ })` severs the murder’s incoming causal edges (specifically the `chain_reaction` edge from `EVT_LADY_PERSUADES`, the `affordance_gate` edge from `OBJ_BLOODY_DAGGERS`, and the `ambient_propagation` edge from `LOC_INVERNESS_CASTLE.tension`) and overrides its realisation to “not-occurred”. Crucially the factual mainline is untouched. A user looking at the version sidebar sees a green factual row with a violet shadow child, both browsable.

C.5 Step 4: causal physics propagation

The propagator topologically sorts the causal sub-graph from the intervened node downward and applies Eq. 1 at every edge. Concretely:

- `EVT_DUNCAN_MURDER  $\rightarrow$  EVT_MALCOLM_FLEES` (`chain_reaction`, $w = 0.85$): the source’s outcome is “ $\emptyset$ ”, so the impact is zero, and `EVT_MALCOLM_FLEES` is marked `blocked_by_intervention`.
- `EVT_DUNCAN_MURDER  $\rightarrow$  ENT_MACBETH.guilt` (`mutation`, $w = 0.7$): blocked. The downstream `state_timeline` delta that would have raised guilt from 0.1 to 0.7 is suppressed; the entity’s reconstructed trait at the banquet remains close to its battlefield baseline.
- `EVT_DUNCAN_MURDER  $\rightarrow$  EVT_BANQUO_SUSPECTS` (`chain_reaction_via_belief`, $w = 0.6$): blocked. `ENT_BANQUO`’s suspicion belief never fires; the `Belief(target=ENT_MACBETH, perceived_state= "Macbeth might be the king’s killer")` is absent from his reconstructed belief set in the shadow branch.
- `EVT_LADY_PERSUADES  $\rightarrow$  ENT_LADY_MACBETH.dominance` (`mutation_social`, $w = 0.5$): not blocked, because the persuasion event itself is upstream of the intervention. Lady Macbeth’s dominance still spikes; her subsequent collapse arc is then re-evaluated against the now-absent causal trigger of guilt.
- Spatial affordance check at every step: a putative shock from `LOC_DUNSINANE_CASTLE` to `ENT_MACDUFF` requires a connected spatial path; if Macduff is in `LOC_ENGLAND` with no open channel, the shock is blocked.

The result is a structured `CausalPhysicsResult{mutations: [...], blocked: [...], hidden_deltas: [...], intervened_nodes: [EVT_DUNCAN_MURDER]}`. The `blocked` list is what makes the engine inspectable: every causal edge that *would have* fired in the factual mainline is recorded with the reason it did not.

C.6 Step 5: narrative-physics scoring

The four structural scorers and the six emotion scorers are now re-evaluated on the shadow branch. The user’s directive (“show what changes for Macduff, Banquo, Malcolm, and the witches”) biases the dramatic-irony scorer toward those four entities. The counterfactual collapses several gaps that defined the factual mainline: the audience no longer sees Banquo’s suspicion gap, no longer sees Macduff’s grief-driven vengeance, and the apparition scene is now etiologically dangling (the witches still prophesy, but Macbeth has no kingship to defend). The mystery score on `EVT_DUNCAN_FOUND_DEAD` drops to zero because the event no longer fires; the suspense at the banquet resets, since Banquo’s ghost depended on the murder. The directive assembler records this delta as a `CandidateScoreReport` per candidate continuation it considers for the banquet rewrite.

C.7 Step 6: directive assembly and the brief

`DirectiveAssembler.evaluate_candidate_events()` now has to choose what to put in the banquet scene of the shadow branch. Candidate events are enumerated by walking unblocked affordances reachable from the shadow-branch state: *Macbeth confesses the prophecy to Banquo*, *Lady Macbeth persists and stages the murder herself*, *Duncan publicly names Macbeth as the new Prince of Cumberland*, *Macbeth retreats to Glamis to think*. Each is forked into its own micro-sandbox and propagated. Plausibility gates drop *Lady Macbeth stages the murder herself* because the daggers' `Affordance(action="kill")` is gated on a wielder with sufficient ruthlessness and the trait-evidence support is low for Lady Macbeth in isolation. The remaining survivors are scored against the directive. The winner is wrapped as a `CreativeBrief`: a list of `must_events`, a list of `must_not_events` ("Macbeth kills Duncan", "Banquo dies"), per-trait envelopes (Macbeth's `guilt` envelope is now $[0.1, 0.3]$ rather than $[0.6, 0.8]$), and a syuzhet window. The brief is the only object handed to the renderer.

C.8 Step 7: constrained rendering

The renderer LLM is invoked with the brief as system prompt plus a small style context (the synopsis's voice settings: synoptic-narration, third-person past tense, sparse density). It writes the banquet as a tense scene of overheard whispers, of Banquo unsuspecting and gracious, of Macbeth conspicuously unable to enjoy his elevation; the daggers stay sheathed; no ghost appears, because Banquo is alive. The renderer is free to choose dialogue, description, and pacing, but cannot invent a causal edge, shift a trait outside its envelope, or open a channel that does not exist. The output is a prose passage, not a graph.

C.9 Step 8: audit, re-extraction, and merge

The auditor runs three passes in parallel. The *causal audit* reverse-engineers the prose into (source, target, modality) triples and matches each against the brief; an LLM-emitted "Banquo's ghost rises behind Macbeth" would be flagged as a *miracle step* (no licensing edge), but in this run nothing of the kind appears. The *abduction audit* runs counterfactual probes against the prose — "if Banquo had suspected the prophecy, would the prose still hold?" — and checks the answers against the brief's belief envelope. The *affective audit* re-runs the four scorers on the prose and compares them to the brief's targets. If any pass fails the `engine_thresholds` gate or the `llm_pass` gate, the renderer is re-invoked with the structured feedback appended, up to `max_correction_retries` (default 3). On convergence, `ingestion` is run again on the prose alone, and the resulting graph is diffed against the brief; the diff becomes a new `VersionRow` with `world_id="shadow"` and `ancestor_id` pointing at the original counterfactual root. The user sees the new version appear in the sidebar in violet, can diff it against the factual mainline, and can `promote_branch` it to canon if desired.

C.10 What this walkthrough demonstrates

Three properties matter. **Inspectability:** every intermediate object — the ego-graph, the `CausalPhysicsResult`, the `CreativeBrief`, the `AuditReport` — is a typed Pydantic value, persisted and visible in the UI. **Locality of LLM authority:** the LLM authors the initial ontology (Step 1), the candidate prose (Step 7), and the audit critique (Step 8); everything else is symbolic computation over typed code. **Branch safety:** the counterfactual is a sibling row, not a mutation; canon is preserved, the shadow is diffable and promotable, and the version DAG records the lineage.

D The authoring user interface

The authoring workspace is a NiceGUI single-page application (`shadow_loom_ui/app.py`) reachable on `http://localhost:7860`. It is organised as a left-hand *version sidebar* and eight cross-linked tabs that share a central `AppState` pub/sub event bus (`shadow_loom_ui/state.py`). The same active-version pointer is mirrored to the database so an MCP client and the UI always see the same tip. This appendix walks through every surface in turn, with a concrete example session drawn from the bundled plot fixtures.

D.1 Version sidebar

The sidebar (`shadow_loom_ui/components/version_sidebar.py`) renders the project's full `VersionRow` DAG as a tree rooted at the original ingestion. Factual rows are green, shadow rows (counterfactual forks under `branch_policy=auto`) are violet. Toolbar actions: *Delete* re-parents children through `db.delete_version`; *Reparent* changes a row's `ancestor_id` (rejected if it would multi-root the tree); *Promote to canon* (shadow rows only) calls `db.promote_branch` to copy a shadow onto a fresh factual row; *Diff against factual* (shadow rows only) opens a structured diff against the current factual head. Clicking any row swaps the active version and resets every tab.

Example. After running the *Macbeth refuses the murder* counterfactual of App. C, the sidebar shows a green chain `ingestion` → `manual_edit` → `pipeline_run` for the factual mainline, with a single violet child branching off the `pipeline_run` row. Right-clicking the violet row offers *Diff against factual* (showing the absent `EVT_DUNCAN_MURDER` and the suppressed downstream deltas) and *Promote to canon* (which would replace the mainline).

D.2 Story tab

The STORY tab (`components/story_tab.py`) is the entry point: a plain-prose textarea plus a *Save & Re-ingest* button. Saving runs the full `pipeline.run_pipeline` and persists the result as a new version with `source="ingestion"`. This tab is intentionally not autosaved: the textarea is prose, not graph, and clicking re-ingest is a deliberate action.

Example. Pasting the public-domain `sample_plots/death_on_the_nile.txt` synopsis and clicking *Save & Re-ingest* produces, after the ingestion fan-out described in App. C.2, a `WorldStateV1` containing `ENT_POIROT`, `ENT_LINNET`, `ENT_JACQUELINE`, `ENT_SIMON`, the `LOC_KARNAK` steamer, and the conspiracy event `EVT_PLAN_HATCHED` with `syuzhet_index` placed late but `fabula_time` placed early — the very gap that the dramatic-irony scorer reads off in App. B.4.

D.3 Explorer tab

The EXPLORER tab (`components/explorer_tab.py`) is a read-only tree view of every node in the active world model, grouped by category, with an inspector panel for the selected node. It is the canonical way to confirm that ingestion has produced the right schema before running queries.

Example. Loading the Macbeth fixture and clicking `ENT_MACBETH` opens an inspector showing each `TraitVector` (`ambition: 0.7, inertia 0.55, evidence strong; guilt: 0.1, inertia 0.25, evidence weak`), each Belief with provenance, the `state_timeline` entries (one per major shock event), and the entity's outgoing causal and relationship edges. The inspector is the principal debugging tool when an extraction agent has produced an unexpected trait or belief.

D.4 World tab

The WORLD tab (`components/world_tab.py`) is a Cytoscape graph visualisation of entities, locations, objects, and edges, with a fabula-time slider beneath. Sliding the cursor through fabula time updates node colours and sizes via `state.set_fabula_cursor` (debounced 120 ms) so that the viewer can watch the social topology evolve.

Example. On *Reservoir Dogs*, scrubbing the slider from fabula $t = 0$ (the diner conversation) through to $t \approx 8000$ (the warehouse stand-off) animates the `ENT_ORANGE` node turning from neutral grey to red as the wound state advances; the spatial graph contracts as characters converge on the warehouse; and the `trust` edges from `ENT_WHITE` to `ENT_ORANGE` thicken. The same scrub on *Romeo and Juliet* shows Romeo's despair edge widening sharply over the final scene.

D.5 Causality tab

The CAUSALITY tab (`components/causality_tab.py`) has three sub-tabs. *Topology* renders a Sankey diagram of causal flow up to the fabula cursor: thicker ribbons for higher-weight edges, colour

banded by `causality_type`. *Evolution* plots trait trajectories per entity over fabula time, with a vertical needle synced to the cursor. *Affective Dashboard* shows gauges and time series for the four structural scorers and the six emotion scorers, computed by `viz_helpers.compute_affective_scores` (which routes through `DirectiveAssembler.compute*_score`). The top twenty entities by event degree are shown by default to keep the plots readable.

Example: Macbeth Evolution sub-tab. Selecting `ENT_MACBETH`, `ENT_LADY_MACBETH`, and `ENT_MACDUFF` and scrubbing through fabula time shows Macbeth’s guilt curve climbing in two steps (after `EVT_DUNCAN_MURDER` and again after `EVT_BANQUO_GHOST`); Lady Macbeth’s ruthlessness curve peaks early and then collapses into the sleepwalking scene; and Macduff’s grief curve has a single step at the slaughter event. Switching to the *Affective Dashboard*, the suspense gauge sits high through Acts II–IV and zeroes out cleanly when the duel begins (Wilmot and Keller, 2020’s suspense–despair boundary).

Example: Death on the Nile Affective Dashboard. The dramatic-irony gauge sits at ~ 0.7 from the moment the conspiracy is revealed to the audience and drops to zero only when Poirot delivers the denouement; the mystery gauge follows the inverse trajectory, peaking on the morning of the murder.

D.6 Reasoning tab

The REASONING tab (`components/reasoning_tab.py`) is where queries are typed and where intervention / counterfactual traces are inspected. The chat box accepts natural-language queries that are parsed into one of the eight typed query objects (`ObservationQuery`, `InterventionQuery`, `CounterfactualQuery`, `DirectiveQuery`, `InterrogateQuery`, `GeneralQuery`, `ManualEditQuery`, `EvaluateQuery`). The result panel uses `reasoning_helpers.py` to format the `CausalPhysicsResult`: each mutation is shown with its (I, ι, Δ) triple, each blocked propagation with the reason (inertia, affordance, spatial), each abduction back-fill with the evidence weight, and any hidden deltas with the syuzhet window in which they will be revealed.

Example. On Macbeth, typing “If Macbeth had told Banquo about the prophecy on the heath, would Banquo have trusted him at the banquet?” parses to a `CounterfactualQuery` with focal entities `{ENT_MACBETH, ENT_BANQUO}`. The trace panel shows the abduction step (Banquo’s `trust` prior is inferred from the battlefield events), the do-step (a new utterance event is inserted on the heath channel), and the propagation step (Banquo’s `suspicion` trait is suppressed at the banquet, but his `prudence` trait *raises* a new caution gate, which the engine reports as a non-trivial counterfactual outcome — not all secrets shared are secrets defused).

D.7 Audit tab

The AUDIT tab (`components/audit_tab.py`) displays the structured `AuditReport` for the most recently rendered scene. The three sections (causal, abductive, affective) each show their loss values and any offending prose spans, highlighted inline. A green badge indicates a passed pass; an amber badge indicates a warning (the prose is acceptable but drifted); a red badge indicates a hard failure that triggered a re-render.

Example. On *Gone Girl*, after generating a continuation of Amy’s diary entry, the causal audit flags a hallucinated “Nick suddenly suspects Amy is alive” span with no licensing causal edge in the brief. The auditor’s structured feedback is displayed alongside the offending span, and the user can either accept the auto-rerun or open the brief in the REASONING tab to widen the envelope.

D.8 Editor tab

The EDITOR tab (`components/editor_tab.py`) is the manual world-model editor. It has three integrated surfaces backed by a single JSON textarea (the source of truth): a *Validate* button that runs the same `_programmatic_validation` as ingestion; a *Structural editor* with one collapsible panel per collection (locations, entities, objects, events, world traits, causal edges, relationship edges, spatial edges, channels); and the raw JSON textarea itself for fields the structural editor does not expose (sparse

trait values, beliefs, ambient state, evidence strengths). The save flow runs Pydantic validation, then `_programmatic_validation`; errors block the save with a dialog listing every issue, warnings present a *Save Anyway* confirmation, and on confirm the world is persisted as a new `VersionRow` with `source="manual_edit"` parented to the current row.

Example. Authoring the `example_worlds/macbeth.py` fixture from scratch, the user adds `LOC_HEATH` via the structural editor (typing “*the heath*” into the ID field auto-prefixes to `LOC_THE_HEATH`, edited to `LOC_HEATH`), then `ENT_MACBETH`, then the witches as a single multi-actor entity, then the prophecy event with `event_type= revelation` and `via_channel_id` pointing at a freshly added `Channel(participants=[ENT_MACBETH, ENT_BANQUO, ENT_WITCHES], medium="speech", intelligibility={...: 1.0})`. After each addition the validator panel updates: the green panel shows the running counts, an amber panel warns about the missing mutation coverage on `ENT_MACBETH.ambition` (no `state_timeline` yet), and a red panel reports the broken `addressee_ids` reference until the witches entity is added.

D.9 Export tab

The EXPORT tab (`components/export_tab.py`) is the egress surface: download the active world as a single `world_state.json`, download the version tree as a structured JSONL file, or copy the active version row ID for use with the MCP server or scripts.

Example. A researcher running cross-fixture comparisons exports each of the sixteen bundled fixtures’ `world_state.json` files, computes the entity-degree distribution and the per-fixture suspense time-series in a notebook, and persists the results without ever loading the UI’s visualisation code.

D.10 Performance discipline

Three small disciplines keep the workspace responsive during long scrubs and rapid edits. (i) Cursor writes are debounced through `state.set_fabula_cursor`, which schedules an emit on a 120 ms tail. (ii) Heavy panels are gated by `is_path_visible(path)` so that off-tab rebuilds are skipped. (iii) `state.spawn_panel_task(panel_id, coro)` cancels any prior task with the same `panel_id`, collapsing rapid scrubs to the most recent render; ECharts panels build the chart once and patch options in place.

D.11 Cross-tab event bus

Six events flow through `AppState`: `PROJECT_LOADED` (emitted by the project picker and the MCP `open_project` tool, listened to by every tab); `WORLD_STATE_CHANGED` (emitted by `load_db_version`, manual save, and pipeline runs, triggering cache invalidation in `viz_helpers`); `VERSION_CHANGED` (sidebar highlight + raw-text reload); `FABULA_CURSOR_CHANGED` and `SYUZHET_CURSOR_CHANGED` (cursor sliders); and `ACTIVE_PATH_CHANGED` (top tab change, gating heavy panels). Adding a new tab is therefore a matter of subscribing to the relevant subset of these events and rendering against the current `AppState`.

D.12 Why a UI matters for a research artefact

The UI is not an optional consumer surface; it is the principal debugging tool for the symbolic layers. The version sidebar makes branch lineage visible; the Causality tab makes the affective scorers’ trajectories visible; the Reasoning tab makes the causal-physics trace visible; and the Audit tab makes the brief-vs-prose mismatch visible. A reviewer or collaborator can load any of the sixteen bundled fixtures, run the counterfactuals discussed in App. B, and inspect every intermediate object without writing a line of code. That inspectability is what the typed-substrate design was meant to buy; the UI is where it is cashed in.